

Assignment –6

Introduction to stochastic process

1. If a random process is described by

$$Z(t) = X_1 \cos \omega_0 t - X_2 \sin \omega_0 t$$

Where X_1 and X_2 are independent normal variates with zero mean and variances σ^2 , then find the expectation and variances of Z . Ans: $E(Z) = 0$, $V(Z) = \sigma^2$

2.

If a random process $\{X(t)\}$ is given by,

$$X(t) = \begin{cases} A; & t \in [0, 1]. \\ 0; & \text{Elsewhere.} \end{cases}$$

Where $A \sim U[-\theta, \theta]$, then find the corresponding autocorrelation function of $R_X(t_1, t_2)$.

$$\text{Ans: } R_X(t_1, t_2) = \frac{\theta^2}{3}$$

3.

If a random process is given by $X(t) = \cos(\omega t + \Theta)$ where $\Theta \sim U[-\pi, \pi]$, then find the mean, autocorrelation, and autocovariance function.

$$\text{Ans: } E(X) = 0, R_X(t_1, t_2) = C_X(t_1, t_2) = \frac{\cos \omega(t_1 - t_2)}{2}$$

4.

If a random process is given by $X(t) = A \cos \omega t$ where, $A \sim U[0, 1]$, find the autocorrelation and autocovariance function.

$$\text{Ans: } R_X(t_1, t_2) = \frac{1}{3} \cos \omega t_1 \cos \omega t_2 : C_X(t_1, t_2) = \frac{1}{12} \cos \omega t_1 \cos \omega t_2$$

5.

If a random process is given by $X(t) = a \cos(bt + Y)$ where, a, b are constants and $Y \sim U[-\pi, \pi]$, then find the autocorrelation function.

$$\text{Ans: } R_X(t_1, t_2) = \frac{a^2}{2} \cos b(t_1 - t_2)$$

6.

If a random process is given by $X(t) = \cos(\omega_0 t + \Theta)$ where, $\Theta \sim U[-\pi, \pi]$, show that the first and second order moments are independent of time t . Moreover, check if Θ is replaced by a constant θ_0 then whether the mean is independent of time or not?

Ans: $E(X) = 0, E(X^2) = \frac{1}{2}, E(X) = \cos(\omega_0 t + \theta_0)$

7.

If a random process is given by $X(t) = a \cos(bt + Y)$ where, a, b are constants and $Y \sim U[0, \pi/2]$, find the autocorrelation function.

Ans: $R_X(t_1, t_2) = \frac{-a^2}{\pi} \sin b(t_1 + t_2) + \frac{a^2}{2} \cos b(t_1 - t_2)$

8.

If a random process is given by $X(t) = A \cos(2\pi t + Y)$, Where A is a constant and Y is discrete random variable with $P(Y = 0) = \frac{1}{2}$ and $P(Y = \pi/2) = \frac{1}{2}$, then find the mean and autocorrelation function.

Ans: $E(X) = \frac{A}{2} \{\cos 2\pi t - \sin 2\pi t\}; R_X(t_1, t_2) = \frac{A^2}{2} \cos 2\pi(t_1 - t_2)$

9.

If a random process is given by $X(t) = A \cos(\omega t + \Theta)$, Where A, ω are constants and $\Theta \sim U[-\pi, \pi]$, then find the autocorrelation function for $Y(t) = X^2(t)$.

Ans: $R_Y(t_1, t_2) = \frac{A^4}{8} [2 + \cos 2\omega(t_1 - t_2)]$

10.

If a random process $\{X(t)\}$ has mean $E[X(t)] = 3$ and the autocorrelation function $R_X(t_1, t_2) = 16 + 4e^{-0.3|t_1 - t_2|}$, then Find the mean, the variance of the random variable $Z = X(8)$.

Ans: $E[X(8)] = V[X(8)] = 4$

11.

If a random process is given by $X(t) = A + Bt$ where A and B are independent random variables with $E(A) = a, E(B) = b$ and $V(a) = \sigma_a^2, V(b) = \sigma_b^2$, then find the mean, autocorrelation and autocovariance.

$E[X(t)] = A + Bt; R_X(t_1, t_2) = (\sigma_a^2 + a^2) + (t_1 + t_2)ab + t_1 t_2 (\sigma_b^2 + b^2)$
 $C_X(t_1, t_2) = \sigma_a^2 + \sigma_b^2 t_1 t_2$